WORKED SOLUTION

STEP II 2014, Question 4

Pure • Integration by Substitution

STEP II 2014, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q4

The key idea

The substitution $u = 1/x$ swaps the two ends of the interval $[1/b, b]$, so it turns the integral I back into an integral over the same range. If the new integrand is the old one with a sign change, we get $I = -I$. If it's something closely related, adding the two forms of I can give an integrand that is much easier to integrate. In the final part the same trick, with $u = a^2/x$ so that $[0, \infty)$ maps to itself, gives an equation that we can solve for the integral we want.

Solution

Part (i)

Call the integral I. With $x = 1/u$ we have $\mathrm{d}x = -\frac{1}{u^2}\,\mathrm{d}u$ and $\ln x = -\ln u$. When $x = 1/b$, $u = b$, and when $x = b$, $u = 1/b$. Also

$$(a^2 + x^2)(a^2x^2 + 1) = \left(a^2 + \frac{1}{u^2}\right)\left(\frac{a^2}{u^2} + 1\right) = \frac{(a^2u^2 + 1)(a^2 + u^2)}{u^4}.$$

So

$$\begin{aligned} I &= \int_b^{1/b} \frac{\frac{1}{u}(-\ln u)}{(a^2u^2 + 1)(a^2 + u^2)/u^4}\left(-\frac{1}{u^2}\right)\mathrm{d}u \\ &= \int_b^{1/b} \frac{u \ln u}{(a^2 + u^2)(a^2u^2 + 1)}\,\mathrm{d}u = -\int_{1/b}^{b} \frac{u \ln u}{(a^2 + u^2)(a^2u^2 + 1)}\,\mathrm{d}u. \end{aligned}$$

The last integral is just I again with the dummy variable renamed. So $I = -I$, which gives $I = 0$, as required. (The integrand is continuous on $[1/b, b]$ because $b > 0$, so I is a finite number and this step is valid.)

Part (ii)

Call this integral I. The same substitution gives $\frac{\mathrm{d}x}{x} = u \cdot (-\frac{1}{u^2})\mathrm{d}u = -\frac{\mathrm{d}u}{u}$, and the limits swap as before:

$$I = \int_b^{1/b} \arctan\Big(\frac{1}{u}\Big)\Big(-\frac{1}{u}\Big)\mathrm{d}u = \int_{1/b}^{b} \frac{\arctan(1/u)}{u}\,\mathrm{d}u.$$

This time the sign doesn't simply flip, so we add this form to the original one:

$$2I = \int_{1/b}^{b} \frac{\arctan u + \arctan(1/u)}{u}\,\mathrm{d}u.$$

Ask yourself What is $\arctan u + \arctan(1/u)$ when $u > 0$?

If $\theta = \arctan u$ with $u > 0$, then $0 < \theta < \frac{\pi}{2}$ and $\tan(\frac{\pi}{2} - \theta) = \cot\theta = 1/u$. Since $\frac{\pi}{2} - \theta$ also lies in $(0, \frac{\pi}{2})$, it is $\arctan(1/u)$. So $\arctan u + \arctan(1/u) = \frac{\pi}{2}$ for all $u > 0$. (For $u < 0$ the sum is $-\frac{\pi}{2}$, but every u in $[1/b, b]$ is positive.) Hence

$$2I = \frac{\pi}{2}\int_{1/b}^{b} \frac{\mathrm{d}u}{u} = \frac{\pi}{2}(\ln b - \ln(1/b)) = \pi \ln b,$$

so $I = \dfrac{\pi \ln b}{2}$, as required. This holds whether $b > 1$ or $0 < b < 1$.

Part (iii)

Let $J = \displaystyle\int_0^{\infty} \frac{\mathrm{d}x}{(a^2 + x^2)^2}$. This converges, since the integrand is continuous and at most $1/x^4$ for large x. We want a substitution that maps $(0, \infty)$ to itself and turns $a^2 + x^2$ into a multiple of $a^2 + u^2$, so we take $u = a^2/x$, i.e. $k = a^2$. Then $x = a^2/u$, $\mathrm{d}x = -\frac{a^2}{u^2}\,\mathrm{d}u$, $x \to 0^+$ corresponds to $u \to \infty$, and

$$a^2 + x^2 = a^2 + \frac{a^4}{u^2} = \frac{a^2(u^2 + a^2)}{u^2}.$$

Therefore

$$J = \int_{\infty}^{0} \frac{u^4}{a^4(a^2 + u^2)^2}\Big(-\frac{a^2}{u^2}\Big)\mathrm{d}u = \frac{1}{a^2}\int_0^{\infty} \frac{u^2}{(a^2 + u^2)^2}\,\mathrm{d}u.$$

Now write $u^2 = (a^2 + u^2) - a^2$ in the numerator:

$$\frac{u^2}{(a^2 + u^2)^2} = \frac{1}{a^2 + u^2} - \frac{a^2}{(a^2 + u^2)^2}.$$

Both pieces have convergent integrals over $(0, \infty)$, so we may split the integral. Using the given result,

$$J = \frac{1}{a^2}\Big(\frac{\pi}{2a} - a^2 J\Big) = \frac{\pi}{2a^3} - J.$$

Hence $2J = \dfrac{\pi}{2a^3}$, so

$$\int_0^\infty \frac{\mathrm{d}x}{(a^2+x^2)^2} = \frac{\pi}{4a^3},$$

as required.

Checks

- In (ii), $b = 1$ gives an interval of zero length, and indeed $\frac{\pi}{2}\ln 1 = 0$. Replacing b by $1/b$ swaps the limits, so the integral should change sign, and $\ln(1/b) = -\ln b$ agrees.
- In (iii), the answer should scale correctly: putting $x = at$ gives $J = a^{-3}\int_0^\infty (1+t^2)^{-2}\,\mathrm{d}t$, so J must be a constant times a^{-3}. With $a = 1$ the substitution $x = \tan\theta$ gives $\int_0^{\pi/2}\cos^2\theta\,\mathrm{d}\theta = \frac{\pi}{4}$, which matches.
- Numerically, the integral in (ii) with $b = 2$ is $1.08879\ldots = \frac{\pi}{2}\ln 2$. The integral in (i) evaluates to 0 (to many decimal places) for several random choices of a and b.

Common slips

- Forgetting that $u = 1/x$ also reverses the limits. The minus sign from $\mathrm{d}x = -\mathrm{d}u/u^2$ and the swap of limits must both be tracked.
- Using $\arctan u + \arctan(1/u) = \frac{\pi}{2}$ without saying that it needs $u > 0$.
- In (iii), getting $\frac{1}{a^2}\int_0^\infty \frac{u^2}{(a^2+u^2)^2}\,\mathrm{d}u$ and then stopping, without spotting that it can be rewritten in terms of J and the given integral.

Beyond STEP

Using a symmetry of the domain to relate an integral to itself is a common trick. It comes back in contour integration and in integrals such as $\int_0^{\pi/2}\ln\sin x\,\mathrm{d}x$. The integrals in (iii) also come from a general family: differentiating $\int_0^\infty(\lambda+x^2)^{-1}\mathrm{d}x = \frac{\pi}{2}\lambda^{-1/2}$ with respect to λ gives the same answer at once ("differentiation under the integral sign"). Real analysis courses explain when steps like this, and the handling of improper integrals, are allowed. See Bartle and Sherbert, *Introduction to Real Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.